AHC VISUAL INTELLIGENCE HACKATHON · 5 SEPT 2026

Real-time video anomaly detection

Can a **small** vision-language model watch live video
and report what matters — fast enough, and without crying wolf?

59.7

out of 100 · was 49.9 one hour before the end

29.8 / 35

on D2 — strongest tier, 4-minute videos localised well

5.5×

faster than real time · 1.83 s per 10 s window

WHAT WE BUILT

Two models. Each blind where the other sees.

Video (sampled by timestamp, fps ranges 1.9 – 30) → **20-second windows** → **Qwen3-VL-8B** (prompted · "what is happening") / **SigLIP2 + GRU head** (frozen encoder · "what, over time") → **Merge + per-difficulty policy** (→ submission)

The VLM finds appearance — fire, smoke, flood, crashes.
The frozen encoder finds duration — loitering, stalled vehicles.
Neither model does both.

MODELS TESTED

Five VLMs, identical test. Two surprises.

Model	Size	Precision	Recall	Verdict
Qwen3-VL-8B-Instruct	**8B**	**0.62**	**0.40**	**Shipped — produced D1 + D2**
Qwen3-VL-4B-Instruct	4B	0.77	0.50	Ties the 8B (differ on 3 of 24)
Cosmos-Reason2-8B	8B	0.57	0.20	Half the recall of a 4B
Qwen3-VL-2B-Instruct	2B	0.30	0.15	Too small
Qwen2.5-VL-7B-Instruct	7B	0.00	0.00	Silent on 20 of 24
SigLIP2 + trained GRU head	**frozen**	**— added +2.4 marks —**		**Shipped**

Cosmos-Reason2-8B tops NVIDIA's own traffic-anomaly leaderboard — and got half the recall of a 4B here. **Benchmark leadership does not transfer.**
SigLIP was marked "rejected" in our own ledger, on theory. It turned out to be one of the most effective approaches on this task. The evidence was there; our theory was wrong.

THE BREAKTHROUGH · FINAL HOUR

+9.8 marks. We changed no model.

Eight prediction sets — three model sizes, five window configs, a fine-tune, targeted prompts — all scored **exactly 8.0** on the long videos.

That invariance was the clue. If changing the model changes nothing, the failure isn't in the model.

Our predicted spans were **1–8 seconds** inside videos of **327–602 seconds**. A 5-second span can never overlap a 100-second event by half — **geometrically impossible, however right the class is.**

49.9 → 56.0

widening spans to 112–138 s.
No new inference. No changed class.

Our classes had been right all along.

→ 59.7

adding SigLIP2, which predicts

loitering

— a class we score 0/6 on and

no VLM proposed all day.

> The evidence was in our own notes from that morning: the sample data held a **125-second** event. We wrote down that these events are long — then spent hours swapping models instead of spans.

HOW WE WORKED

Three AI agents, one repository.

Agent	Role	Owned
Coordinator	Research, strategy, every GPU run, every commit	inference lane, Modal driver
Builder	Data pipeline, synthetic video generation, training, scoring	data lane, tests
Comms	Live demo, slides, process record	demo, docs

- **Ownership by directory.** No agent edits another's files. We learned it the hard way — two agents built the same four modules before the rule existed.
- **One committer**, so the history stayed readable on a graded repo.
- **A written ledger, not chat.** Every technique across 7 tiers marked done / running / rejected — **with the reason**. An agent could crash, restart, and resume from files. It happened twice.
- **They corrected each other.** One caught a scorer using first-fit instead of best-IoU matching. One caught a run that exited "successfully" having done 6 of 34 videos.

WHAT WE LEARNED

Ten failures in one day. None of them crashed.

- **ffmpeg's seek silently returns more than you ask for.** A 468 s video rendered as 241 s — every timestamp after it was wrong.
- **The fix for one bug caused another.** We stopped one bad video killing a 34-video run; it then wrote 10 failures to disk as confident, plausible "nothing here".
- **A model reported 100% validation accuracy.** It had 7 validation samples, because an upload had died unnoticed.
- **We deleted 125 videos we had already rendered** — a third of our synthetic training data — rather than train on labels we could not verify.
- **We retracted our own headline finding** before it shipped. The retraction was load-bearing: keeping it meant shipping the weaker model.

> In a seven-hour build, **every failure that mattered was silent.**
> So we validated intermediate artifacts — not outputs.